



Dynamic, real-time HVAC optimisation - delivering energy savings automatically.

Local Government Civic Office

Dynamic HVAC optimisation alongside wider decarbonisation

£37.5k138 MWh8,500 m²		
Calculated first-year net saving	First-year energy reduction against expected profile	Civic office existing BMS retained

Measured against the expected post-decarbonisation energy profile | Wider low-carbon measures already underway | Existing controls retained | Deployment began February 2022

17.9% Electricity below expected consumption 222,278 kWh below the expected first-year profile	8.0% Total energy below expected consumption 137,701 kWh net reduction across electricity and gas
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Executive summary

A UK local authority was implementing a wider building decarbonisation programme across its estate, including heat pumps, solar generation, battery storage and Building Management System improvements. At an 8,500 m² civic office, EcoAI's optimisation technology was integrated with the existing BMS to improve HVAC operation while retaining the building's installed controls infrastructure.

The challenge was not simply to compare one year with another. The wider carbon-reduction measures were expected to materially change the building's energy profile, with electricity consumption forecast to increase by 35% and gas consumption forecast to reduce by 70%. Performance was therefore assessed against the expected post-project profile.

Across the first 12 months, electricity consumption was 222,278 kWh below expected. Gas consumption was 84,577 kWh above the expected level, giving a net energy reduction of 137,701 kWh. At the rates used in the project analysis, this equated to a calculated net saving of £37,474.87.

The project shows how dynamic HVAC optimisation can work alongside wider decarbonisation investment, helping improve operational performance without requiring the existing controls platform to be replaced.

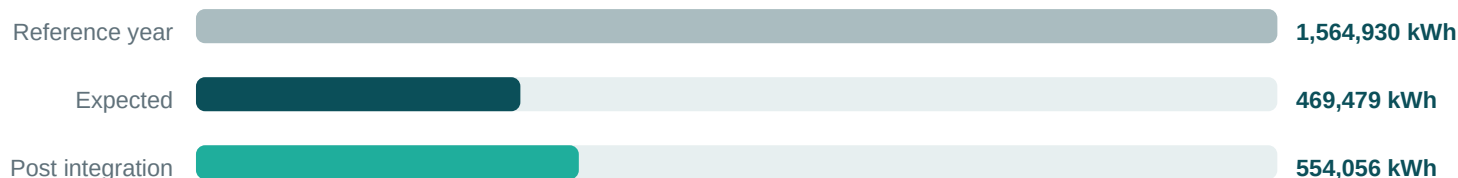
Measured performance vs expected profile

The first-year analysis compared actual post-integration performance with the energy profile expected after the site's wider decarbonisation measures. This matters because the building was undergoing a major shift from gas towards electricity, making a simple historic year-on-year comparison potentially misleading.

Electricity consumption



Gas / heating consumption



Expected profile: the wider project anticipated a 35% increase in electricity use and a 70% reduction in gas use as other carbon-reduction measures were implemented.

Result	First-year position
Electricity	Expected: 1,243,324 kWh Actual: 1,021,046 kWh Saving: 222,278 kWh
Gas / heating	Expected: 469,479 kWh Actual: 554,056 kWh Variance: +84,577 kWh
Net energy position	137,701 kWh reduction against expected consumption
Electricity cost	Expected: £246,802.80 Actual: £202,680.07 Saving: £44,122.73
Gas / heating cost	Expected: £36,901.65 Actual: £43,549.51 Variance: +£6,647.86
Calculated net saving	£37,474.87

Verification note. The figures reproduce the first-year project analysis. The calculated £37.5k saving is the difference between the expected and actual energy costs used in that analysis.

Background and deployment

The local authority had invested significantly in energy-efficiency and carbon-reduction measures across a portfolio of public buildings. The civic office combined an existing BMS with a range of mechanical systems and was selected as a practical location to improve operational performance alongside the wider programme.

EcoAI's optimisation technology was integrated with the existing Building Management System. It coordinated heating, ventilation and cooling operation using building conditions together with local weather and forecast information, while making greater use of the building's thermal behaviour.

The existing BMS and plant remained in place. During operation, the project also helped identify systems requiring attention, providing additional operational insight alongside the energy optimisation.

Working alongside decarbonisation

This project is particularly relevant to local government because the building's energy profile was changing as other measures were introduced. Heat electrification and other carbon-reduction projects can shift demand between fuels, which means optimisation performance should be assessed against the expected future operating profile, not just a historic baseline.

In this case, the expected first-year position allowed for substantially lower gas use but materially higher electricity demand. EcoAI's contribution was then evaluated against that adjusted profile, separating the optimisation effect from the wider programme as far as the available data allowed.

CLIENT PERSPECTIVE

"Having completed many of the easy wins including behavioural changes and capital investment, we have been on the lookout for innovative technologies particularly around a buildings BMS system which can be overlooked when it comes to energy savings. Together with the benefits to the occupants, EcoAI has the potential to save a significant amount of energy both across electricity and gas within a short payback. The bonus being that the savings are guaranteed over the lifetime of the payback."

Energy Resource Manager, UK Local Authority

What the project demonstrates

EXISTING BMS

Optimisation can be introduced through existing controls infrastructure rather than requiring a wholesale replacement project.

WIDER PROGRAMME

Operational optimisation can complement heat pumps, on-site generation, storage and other decarbonisation measures.

MEASURED VALUE

The business case can be evaluated against an adjusted expected profile when other projects are changing the building's energy mix.

Operational value beyond the headline saving

The project also improved visibility of building-system performance. The deployment highlighted systems requiring attention, helping the facilities team focus engineering effort where it could make the greatest difference.

This is an important part of EcoAI's role: the optimisation layer supports the existing engineering team and BMS rather than replacing them, while the data generated can help surface operational issues and opportunities for further improvement.

A practical local-government reference for optimising existing buildings as estates decarbonise.

EcoAI works through the existing Building Management System to improve HVAC operation dynamically, supporting engineering teams while retaining the installed controls infrastructure.